SIMATS ENGINEERING
ASSESSMENT TOOL 2

19/20

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I J. Kirthiga. 192421265 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: J. Kirthiga

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ____ → Transport → Network → ____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔→ Data Link ✔→ Network ✖→ Transport ✖→ Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

NAME:KIRTHIGA J

REG.NO : 192421265

COURSE CODE: CSA0702

SLOT:B

COURSE NAME : COMPUTER NETWORKS

Q1 A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

The given sequence represents the layers of the OSI Model.

The complete order of layers is:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Hence, the missing layers are:

- Session Layer
- Data Link Layer

Session Layer (Layer 5)

The Session Layer is responsible for establishing, managing, and terminating communication sessions between two devices. It ensures that a proper connection is maintained throughout the data exchange process.

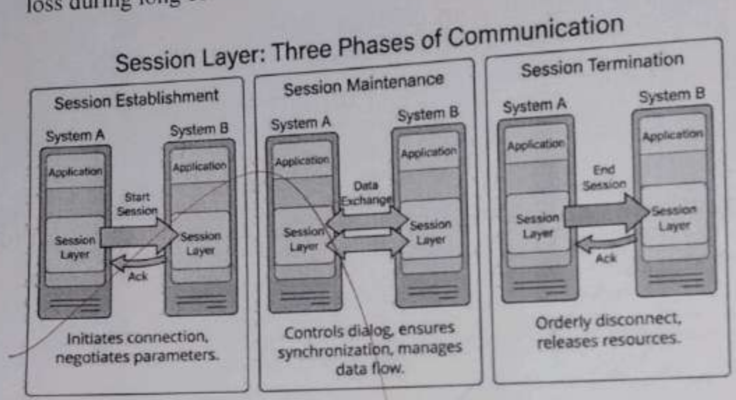
Functions:

- Establishes, maintains, and terminates sessions between applications
- Controls the dialogue (decides which device transmits data at a given time)

- Provides synchronization using checkpoints to recover data in case of failure

Importance for Reliable Communication:

The Session Layer ensures that communication between devices remains stable and organized. If a failure occurs during transmission, the session can resume from the last checkpoint instead of restarting completely. This improves efficiency and prevents data loss during long communications.



Data Link Layer (Layer 2)

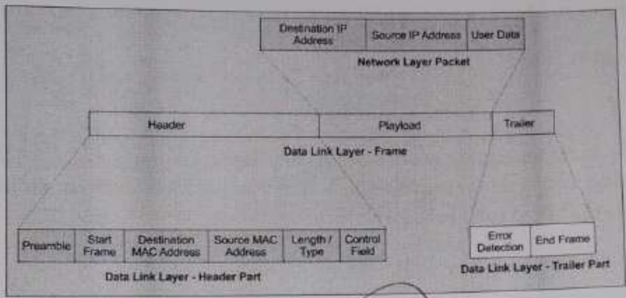
The Data Link Layer is responsible for reliable node-to-node data transfer. It converts raw bits from the Physical Layer into frames and ensures error-free transmission between directly connected devices.

Functions:

- Frames the data into structured units called frames
- Provides error detection and correction
- Uses MAC addresses for device identification
- Controls data flow between sender and receiver

Importance for Reliable Communication:

The Data Link Layer ensures that data is transmitted accurately between adjacent devices. It detects errors and may retransmit corrupted frames, thus preventing incorrect data delivery. It also manages data flow to avoid congestion and data loss.



Frame Structure - Data Link Layer

Literature Insights & Evidence:

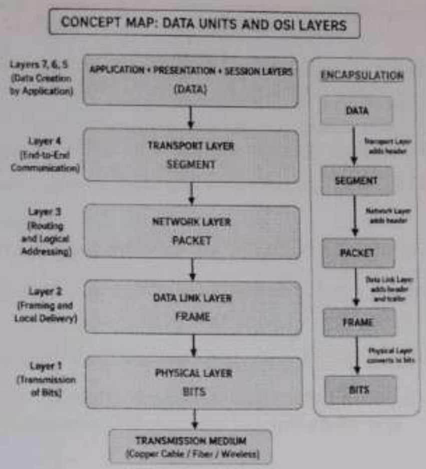
- Shannon (1948): Introduced the concept of information theory and reliable data transmission over noisy channels.
- Tanenbaum (2021): Explains layered network architecture and the functions of each OSI layer.
- Kurose & Ross (2022): Discuss principles of data communication, including error control and flow management.
- Stallings (2020): Describes protocols and mechanisms that ensure reliable and secure network communication.

Q2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Analysis

The OSI (Open Systems Interconnection) model divides the communication process into seven layers. As data moves from the sender to the receiver, it is transformed into different Protocol Data Units (PDUs). This process is called encapsulation.

1. Transport Layer → Segment
   - The Transport Layer converts application data into Segments.
   - It adds source and destination port numbers and sequence information.
   - This ensures reliable end-to-end communication, error recovery, and proper ordering of data.
2. Network Layer → Packet
   - The Network Layer converts each segment into a Packet.
   - It adds source and destination IP addresses.
   - Packets are routed through different networks to reach the destination.
3. Data Link Layer → Frame
   - The Data Link Layer converts the packet into a Frame.
   - It adds source and destination MAC addresses, along with error detection information (CRC).
   - Frames ensure reliable communication within the local network.
4. Physical Layer → Bits
   - The Physical Layer converts the frame into Bits (0s and 1s).
   - These bits are transmitted as electrical, optical, or radio signals through the communication medium.



How the Transformation Supports Transmission

- Segment provides reliable communication and data sequencing.
- Packet enables routing between different networks using IP addresses.
- Frame provides local delivery and error detection using MAC addresses.
- Bits represent the actual binary signals transmitted through wired or wireless media.

Each layer adds its own header (and the Data Link Layer also adds a trailer). This process is known as encapsulation. At the receiving end, the headers are removed layer by layer through decapsulation, allowing the original data to be reconstructed accurately.

Literature Insights & Evidence:

- **ISO (1984):** Introduced the **OSI model** with seven communication layers.
- **Tanenbaum (2021):** Explains **OSI layers** and data unit transformation (Segment, Packet, Frame, Bits).
- **Kurose & Ross (2022):** Describe **encapsulation** and reliable data transmission.
- **Stallings (2020):** Discuss network protocols, error control, and secure communication.

Q3. Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

The concept map shows how the Data Link Layer and the Network Layer work together to ensure complete data delivery across networks.

Data Link Layer → MAC Address → Local Delivery:
The Data Link Layer uses MAC (Media Access Control) addresses to deliver frames between devices on the same local network (LAN).

Network Layer → IP Address → Global Delivery:
The Network Layer uses IP addresses to identify the source and destination across different networks and routes packets to the correct destination.

How They Work Together

The Data Link Layer and the Network Layer complement each other during data transmission.

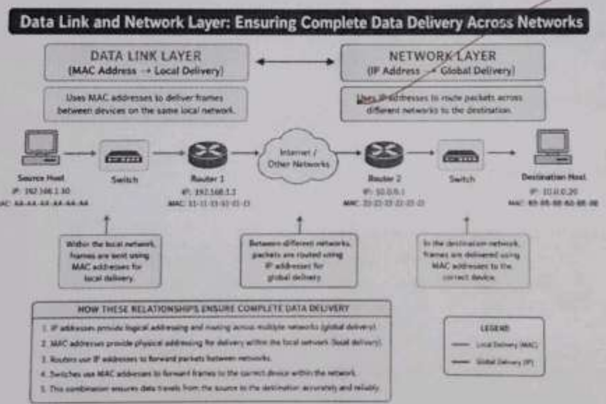
1. The sender creates data and assigns the destination **IP address**.
2. The Network Layer routes the packet through different routers based on the IP address.
3. At each network, the Data Link Layer encapsulates the packet into a frame and uses the destination **MAC address** for local delivery.
4. Switches forward the frame using MAC addresses.
5. This process repeats at every network until the packet reaches the destination.
6. Finally, the destination device receives the data successfully.

MAC Address	IP Address
Used by the Data Link Layer	Used by the Network Layer
Provides local delivery	Provides global delivery

MAC Address	IP Address
Used by switches	Used by routers
Identifies a device on a LAN	Identifies a device across networks

Advantages

- Ensures accurate local communication within a LAN.
- Enables communication across multiple interconnected networks.
- Supports efficient routing and switching.
- Improves reliability and reduces transmission errors.
- Allows seamless Internet communication.



Literature Insights & Evidence:

- **ISO (1984):** Introduced the **OSI model** for network communication.
- **Tanenbaum (2021):** Explains **MAC** for local delivery and **IP** for global delivery.
- **Kurose & Ross (2022):** Describe **switching, routing, and data delivery**.
- **Stallings (2020):** Discusses **network addressing and communication protocols**.

Q4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

Error control is an important function in computer networks that ensures data is transmitted accurately. The Data Link Layer and the Transport Layer work together to detect and correct errors, improving the reliability of communication.

1. Data Link Layer → Error Detection

The Data Link Layer (Layer 2) is responsible for detecting errors during data transmission within the local network.

- Uses CRC (Cyclic Redundancy Check) and checksums.
- Detects corrupted or damaged frames.
- Discards frames containing errors.
- Prevents incorrect data from moving to higher layers.

2. Transport Layer → Error Correction

The Transport Layer (Layer 4) ensures reliable end-to-end communication.

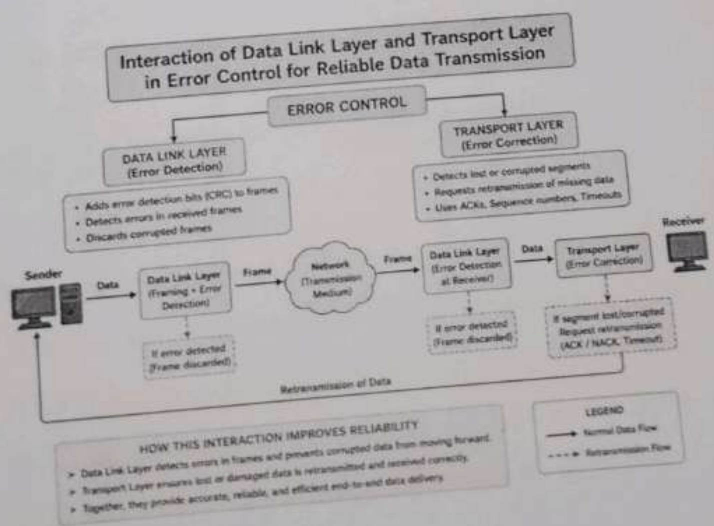
- Detects missing or lost segments.
- Requests retransmission of lost or corrupted data.
- Uses sequence numbers, acknowledgments (ACKs), and timeouts.
- Ensures data reaches the destination in the correct order.

Interaction Between the Layers

1. The Data Link Layer first detects transmission errors in frames.
2. Only error-free frames are passed to the Network and Transport layers.
3. If data is lost or damaged during transmission, the Transport Layer identifies the missing segment.
4. The Transport Layer requests retransmission and restores the original data.
5. Together, these layers ensure accurate and reliable data delivery.

Advantages

- Detects transmission errors quickly.
- Corrects lost or damaged data.
- Improves communication reliability.
- Reduces data corruption.
- Ensures successful end-to-end delivery.



Literature Insights & Evidence:

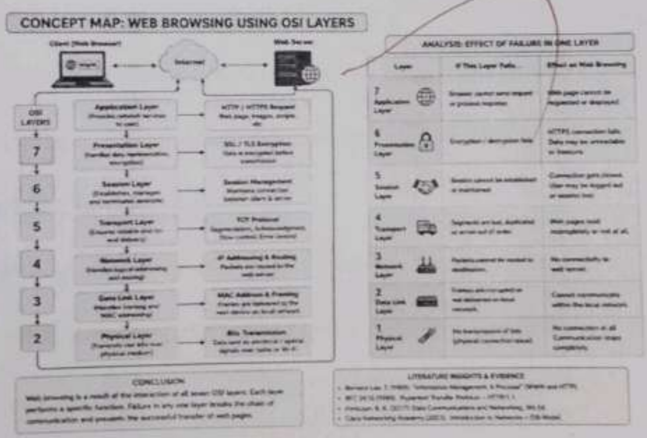
- Cerf & Kahn (1974): Proposed the TCP protocol for reliable end-to-end communication.
- IEEE (2022): Defines Data Link Layer standards for frame transmission and error detection.
- RFC 793 (1981): Specifies TCP error recovery, acknowledgments, and retransmission mechanisms.
- Cisco Networking Academy (2023): Explains how the Data Link and Transport layers work together to improve transmission reliability.

Q5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Analysis

Web browsing uses all seven OSI layers to transfer data between a browser and a web server.

- **Application Layer:** Sends and receives web pages using HTTP/HTTPS.
- **Presentation Layer:** Encrypts and decrypts data using SSL/TLS.
- **Session Layer:** Establishes and maintains the connection.
- **Transport Layer:** Uses TCP to ensure reliable and ordered delivery.
- **Network Layer:** Uses IP addresses to route packets across networks.
- **Data Link Layer:** Uses MAC addresses for local frame delivery.
- **Physical Layer:** Transmits bits through wired or wireless media.



- Effect of Failure in One Layer
  - If the Application Layer fails, the browser cannot request or display web pages.
  - If the Presentation Layer fails, encrypted communication (HTTPS) cannot be established.
  - If the Session Layer fails, the connection between client and server is lost.
  - If the Transport Layer fails, data may be lost, duplicated, or arrive out of order.
  - If the Network Layer fails, packets cannot reach the destination.

- If the Data Link Layer fails, frames cannot be delivered within the local network.
- If the Physical Layer fails, no bits are transmitted, so communication stops completely.

Conclusion

All OSI layers work together during web browsing. A failure in any one layer interrupts communication, preventing the user from successfully accessing web pages.

Literature Insights & Evidence:

- Berners-Lee (1989): Introduced the orld Wide Web and HTTP for web communication.
- RFC 2616 (1999): Defines the HTTP protocol used for web browsing.
- Cisco Networking Academy (2023): Explains how OSI layers support web communication.
- Forouzan (2017): Describes the role of each OSI layer in reliable data transmission.

Q6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

Analyze the problem and explain how the concept map helps in troubleshooting.

Analysis

The concept map shows that the **Physical Layer** and **Data Link Layer** are functioning correctly, but the **Network Layer** has failed. Since the higher layers depend on the Network Layer, the **Transport** and **Application** layers also stop working.

Layer-wise Explanation

1. Physical Layer

- The physical connection (cables, Wi-Fi, signals) is working properly.
- Devices are connected successfully.

2. Data Link Layer

- Frames are transmitted correctly within the local network.
- MAC addresses and switches are functioning normally.

3. Network Layer

- IP routing has failed.
- Possible causes:
  - Incorrect IP address
  - Router failure
  - Wrong subnet mask

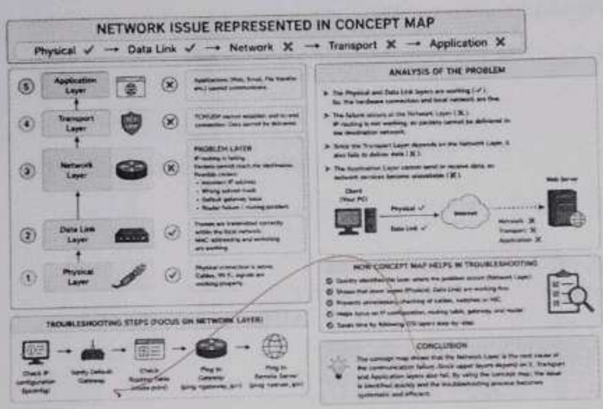
   - Gateway or routing problem

4. Transport Layer

- TCP/UDP cannot establish communication because packets cannot reach the destination.

5. Application Layer

- Applications such as web browsers, email, or file transfer cannot communicate with remote servers.



How the Concept Map Helps in Troubleshooting

1. It identifies the exact layer where the problem occurs.
2. It shows that lower layers are working correctly.
3. It prevents unnecessary checking of cables or switches.
4. It helps administrators focus on **IP configuration, routing tables, gateways, and routers**.
5. It reduces troubleshooting time by following the OSI layers step by step.

Literature Insights & Evidence:

- Cisco Networking Academy (2023): Recommends using the OSI model for systematic network troubleshooting.

- Forouzan (2017): Explains that failures in lower layers affect all dependent upper layers.
- Peterson & Davie (2021): Describe how routing failures at the Network Layer interrupt end-to-end communication.
- RFC 1122 (1989): Defines layered communication and dependency between network protocol layers.